

# Deterministic Projection, Diffusive Limits, and Knee-Like Threshold Transitions

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## Abstract

A finite-dimensional deterministic system can produce a stochastic effective law when only a non-injective observable is retained and hidden initial conditions are sampled from an invariant ensemble. We give an explicit invertible example. Its fast subsystem is the hyperbolic toral automorphism generated by  $A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$ , its slow coordinate is an affine contraction driven by the centered observable  $\phi(y) = \sin(2\pi y_1)$ , and the observed map forgets the torus coordinate. Standard deterministic homogenization then gives an affine Ornstein–Uhlenbeck limit with exact asymptotic variance  $\sigma^2 = 1/2$ . At any fixed readout time, the limiting probability of a positive outcome is an exact Gaussian sigmoid. We compute its midpoint, maximal slope, and every quantile width, and prove uniform convergence of the finite deterministic readout curves on compact protocol intervals. The construction is an existence theorem for deterministic microscopic realizations of diffusive threshold laws. It does not derive intrinsic randomness, a first-passage law, quantum collapse, the Born rule, or an MTT-selected physical protocol.

## Version 2 Revision Note

**Supersedes** *Deterministic Projection, Diffusive Limits, and Knee-Like Threshold Transitions*, version 1.

**Reason** The previous model used the doubling map while calling the full dynamics invertible, reversed the drift geometry at the proposed exit boundary, changed from a stability-crossing model to a different affine-readout model, and then identified an endpoint probability with a first-passage probability. Its comparison with fixed Markov generators was also stronger than the construction justified.

**Resolution** This version uses one smooth invertible hyperbolic toral map throughout, declares the invariant initial-condition ensemble that supplies probability, computes the diffusion coefficient exactly, and treats one fixed-time threshold observable consistently from the finite deterministic model to its Ornstein–Uhlenbeck limit. Endpoint and first-passage events are separated, and the Markov comparison is stated as a microscopic realization result rather than a no-go theorem.

**Retained result** A smooth invertible finite-dimensional skew product can have a non-injective readout whose ensemble law converges to a diffusion and whose protocol-dependent threshold curve has a finite, exactly computable width.

**Remaining boundary** The toy parameters and protocol are not selected by MTT geometry; no physical measurement interaction, Born weights, first-passage theorem, or empirical threshold is derived here.

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## 1 Question, picture, and scope

Suppose a complete state is  $(x, y)$ , but an observer records only  $x$ . Even when the update of  $(x, y)$  is deterministic and invertible, the next value of  $x$  is not determined by the current value of  $x$  alone: different hidden values  $y$  lie in the same observational fiber. If an ensemble of hidden initial conditions is specified, the observed coordinate therefore has a probability law.

This elementary point must be stated carefully. Projection does not make an individual trajectory random. It makes the retained coordinate *statistically incomplete* relative to an ensemble on the discarded coordinates. A diffusion can then appear in a scale-separation limit when the hidden deterministic motion is sufficiently chaotic. Deterministic homogenization makes that statement rigorous [4, 2, 1].

The purpose of this paper is to put the mechanism in one small model where four facts can be checked separately:

1. the full finite-step map is a smooth diffeomorphism;
2. the observational map is non-injective;
3. the projected ensemble law has a rigorous diffusion limit; and
4. the limiting fixed-time threshold curve, including its width, is explicit.

The model is not proposed as a quantum measurement theory. In particular, a positive-coordinate readout is only a binary toy observable. It is useful because it isolates what projection plus deterministic homogenization can prove without importing claims about collapse, first passage, or physical probabilities.

## 2 An invertible deterministic source model

### 2.1 Fast carrier and observable

Let

$$\mathbb{T}^2 = \mathbb{R}^2 / \mathbb{Z}^2, \quad A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix},$$

and let

$$T(y) = Ay \pmod{\mathbb{Z}^2}.$$

Since  $\det A = 1$ ,  $T$  is an invertible smooth map of  $\mathbb{T}^2$ . Its eigenvalues are  $(3 \pm \sqrt{5})/2$ , so neither lies on the unit circle. Thus  $T$  is a hyperbolic toral automorphism. It preserves normalized Lebesgue measure  $\mu$ , is mixing, and satisfies an invariance principle for sufficiently regular centered observables [3].

Choose

$$\phi(y) = \sin(2\pi y_1).$$

This observable is smooth and centered:

$$\int_{\mathbb{T}^2} \phi \, d\mu = 0.$$

### 2.2 Slow coordinate and protocol

Fix constants

$$\gamma > 0, \quad \alpha > 0, \quad s_0 \in \mathbb{R},$$

and choose  $0 < \varepsilon < \gamma^{-1}$ . Set

$$r_\varepsilon = 1 - \varepsilon\gamma \in (0, 1).$$

For a protocol coordinate  $s \in \mathbb{R}$ , define

$$F_{\varepsilon,s}(x, y) = (r_\varepsilon x + \varepsilon\alpha(s - s_0) + \sqrt{\varepsilon}\phi(y), T(y)) \tag{2.1}$$

on  $\mathcal{M} = \mathbb{R} \times \mathbb{T}^2$ . The retained observable is

$$P : \mathcal{M} \longrightarrow \mathbb{R}, \quad P(x, y) = x.$$

It is non-injective because every fiber  $P^{-1}(x)$  contains a full copy of  $\mathbb{T}^2$ .

**Proposition 2.1** (The source map is a diffeomorphism). *For every admissible  $\varepsilon$  and every  $s$ , the map  $F_{\varepsilon,s}$  in (2.1) is a smooth diffeomorphism of  $\mathbb{R} \times \mathbb{T}^2$ .*

*Proof.* Write  $(x', y') = F_{\varepsilon,s}(x, y)$ . Since  $T$  is invertible and  $r_\varepsilon \neq 0$ ,

$$y = T^{-1}(y'), \quad x = \frac{x' - \varepsilon\alpha(s - s_0) - \sqrt{\varepsilon}\phi(T^{-1}y')}{r_\varepsilon}.$$

This is a smooth inverse. □

## 2.3 Where probability enters

Fix  $x_0 \in \mathbb{R}$  and sample only the hidden initial condition,

$$y_0 \sim \mu.$$

Every sampled pair  $(x_0, y_0)$  then follows a deterministic orbit. The symbol  $\mathbb{P}$  below means the pushforward of  $\mu$  through this deterministic evolution; it is not an additional random forcing at finite  $\varepsilon$ .

Let  $(x_n, y_n) = F_{\varepsilon, s}^n(x_0, y_0)$ . Direct iteration gives

$$\begin{aligned} x_n = & r_\varepsilon^n x_0 + \frac{\alpha(s - s_0)}{\gamma} (1 - r_\varepsilon^n) \\ & + \sqrt{\varepsilon} \sum_{j=0}^{n-1} r_\varepsilon^{n-1-j} \phi(T^j y_0). \end{aligned} \quad (2.2)$$

The first line is deterministic. The second is a weighted deterministic Birkhoff sum whose ensemble law is induced by  $y_0 \sim \mu$ .

## 3 Diffusive limit with an exact coefficient

### 3.1 The fast variance

The Green–Kubo variance of  $\phi$  is

$$\sigma^2 = \int_{\mathbb{T}^2} \phi^2 d\mu + 2 \sum_{k=1}^{\infty} \int_{\mathbb{T}^2} \phi(y) \phi(T^k y) d\mu(y). \quad (3.1)$$

**Lemma 3.1** (Exact asymptotic variance). *For the chosen  $T$  and  $\phi$ ,*

$$\sigma^2 = \frac{1}{2}.$$

*Proof.* The zeroth correlation is  $\int_{\mathbb{T}^2} \sin^2(2\pi y_1) d\mu = 1/2$ . The Fourier frequencies of  $\phi$  are  $\pm e_1$ . The frequencies of  $\phi \circ T^k$  are  $\pm(A^\top)^k e_1$ . Hyperbolicity implies  $(A^\top)^k e_1 \neq \pm e_1$  for every  $k \geq 1$ . Orthogonality of distinct torus characters therefore makes every positive-lag correlation in (3.1) vanish.  $\square$

### 3.2 Imported homogenization theorem and specialization

For  $t \geq 0$ , let  $X_s^\varepsilon(t)$  be the piecewise-linear interpolation of  $x_n$  at times  $t_n = n\varepsilon$ . The discrete deterministic-homogenization theorems apply to recurrences with increments of order  $\varepsilon$  in the drift and  $\sqrt{\varepsilon}$  in a centered chaotic observable [2, 1]. The following is their specialization to (2.1); the imported theorem, rather than MTT, supplies the analytic convergence.

**Theorem 3.2** (Projected diffusion limit). *Fix  $S, T_0 > 0$ . For every fixed  $s \in [-S, S]$ ,*

$$X_s^\varepsilon \implies X_s \quad \text{in } C([0, T_0]) \quad \text{as } \varepsilon \downarrow 0,$$

where  $X_s$  is the unique solution of

$$dX_s(t) = (\alpha(s - s_0) - \gamma X_s(t)) dt + \frac{1}{\sqrt{2}} dW(t), \quad X_s(0) = x_0. \quad (3.2)$$

The convergence is in law with respect to  $y_0 \sim \mu$ .

*Proof.* The hyperbolic toral automorphism and smooth centered observable satisfy the weak invariance principle. The slow drift is globally Lipschitz. The additive-noise case of the cited discrete homogenization theorem therefore gives the limiting affine Ornstein–Uhlenbeck equation. Lemma 3.1 supplies the diffusion coefficient.  $\square$

**What the limit says.** At finite  $\varepsilon$ , no Brownian path exists in the source model. Brownian motion describes the limiting law of many hidden deterministic histories when the fast and slow scales separate. The limit is an effective statistical description, not an ontological change in the source dynamics.

## 4 The exact fixed-time threshold curve

Fix a readout time  $t > 0$  and define

$$q(s, t) = \mathbb{P}(X_s(t) > 0).$$

This is an endpoint event. It is not the first-passage event  $\{\sup_{u \leq t} X_s(u) \geq 0\}$ .

The solution of (3.2) is Gaussian. With  $\sigma^2 = 1/2$ , its mean and variance are

$$m(s, t) = x_0 e^{-\gamma t} + \frac{\alpha(s - s_0)}{\gamma} (1 - e^{-\gamma t}), \quad (4.1)$$

$$v(t) = \frac{1}{4\gamma} (1 - e^{-2\gamma t}). \quad (4.2)$$

Let  $\Phi$  and  $\varphi_N$  denote the standard normal distribution and density.

**Theorem 4.1** (Exact endpoint threshold law). *For every  $t > 0$ ,*

$$q(s, t) = \Phi\left(\frac{m(s, t)}{\sqrt{v(t)}}\right). \quad (4.3)$$

*It is strictly increasing in  $s$ , and its unique midpoint is*

$$s_*(t) = s_0 - \frac{\gamma x_0 e^{-\gamma t}}{\alpha(1 - e^{-\gamma t})}. \quad (4.4)$$

*Its derivative is*

$$\partial_s q(s, t) = \varphi_N\left(\frac{m(s, t)}{\sqrt{v(t)}}\right) \frac{\alpha(1 - e^{-\gamma t})}{\gamma \sqrt{v(t)}}, \quad (4.5)$$

*and the maximum occurs at  $s = s_*(t)$ .*

*For  $0 < p_1 < p_2 < 1$ , let  $z_p = \Phi^{-1}(p)$  and let  $s_p(t)$  satisfy  $q(s_p(t), t) = p$ . Then the exact  $p_1$ -to- $p_2$  width is*

$$s_{p_2}(t) - s_{p_1}(t) = (z_{p_2} - z_{p_1}) \frac{\gamma \sqrt{v(t)}}{\alpha(1 - e^{-\gamma t})}. \quad (4.6)$$

*Proof.* Equation (4.3) follows from the Gaussian law with (4.1)–(4.2). The coefficient of  $s$  in  $m$  is positive, giving strict monotonicity. Solving  $m = 0$  gives (4.4). Differentiation gives (4.5); the standard normal density has its unique maximum at zero. Finally, solving  $m(s_p, t)/\sqrt{v(t)} = z_p$  and subtracting the two solutions gives (4.6).  $\square$

For example, the 10–90 width is obtained by taking  $p_1 = 0.1$  and  $p_2 = 0.9$ . It is positive at every finite diffusion strength and tends to zero only in a zero-noise scaling. Thus “knee-like” means a rapid but continuous sigmoid, not a discontinuity or a phase transition.

## 5 Return to the finite deterministic model

Define the finite-system endpoint curve at  $n_\varepsilon(t) = \lfloor t/\varepsilon \rfloor$  by

$$q^\varepsilon(s, t) = \mu \left\{ y_0 \in \mathbb{T}^2 : x_{n_\varepsilon(t)}(s, y_0) > 0 \right\}.$$

Equation (2.2) decomposes the endpoint into

$$x_{n_\varepsilon(t)} = m_\varepsilon(s, t) + Z_\varepsilon(t),$$

where

$$m_\varepsilon(s, t) = r_\varepsilon^{n_\varepsilon(t)} x_0 + \frac{\alpha(s - s_0)}{\gamma} (1 - r_\varepsilon^{n_\varepsilon(t)}),$$

$$Z_\varepsilon(t) = \sqrt{\varepsilon} \sum_{j=0}^{n_\varepsilon(t)-1} r_\varepsilon^{n_\varepsilon(t)-1-j} \phi(T^j y_0).$$

The random variable  $Z_\varepsilon(t)$  is simply a deterministic function of the hidden initial condition.

**Proposition 5.1** (Uniform deterministic readout convergence). *For every  $t > 0$  and every compact protocol interval  $K \subset \mathbb{R}$ ,*

$$\sup_{s \in K} |q^\varepsilon(s, t) - q(s, t)| \longrightarrow 0 \quad (\varepsilon \downarrow 0).$$

*Proof.* Theorem 3.2 implies  $Z_\varepsilon(t) \Rightarrow N(0, v(t))$ . Because the limiting distribution function is continuous, weak convergence of the one-dimensional laws is uniform in the threshold argument. Also  $m_\varepsilon(s, t) \rightarrow m(s, t)$  uniformly for  $s$  in compact sets, since  $r_\varepsilon^{\lfloor t/\varepsilon \rfloor} \rightarrow e^{-\gamma t}$ . Writing each readout probability as the tail distribution of  $Z_\varepsilon(t)$  at  $-m_\varepsilon(s, t)$  proves the claim.  $\square$

**Corollary 5.2** (Convergence of finite-system quantiles). *Fix  $t > 0$  and  $p \in (0, 1)$ . Define*

$$s_p^\varepsilon(t) = \inf\{s : q^\varepsilon(s, t) \geq p\}.$$

*Then*

$$s_p^\varepsilon(t) \longrightarrow s_p(t),$$

*provided the infimum is taken in a compact interval containing the unique limiting quantile.*

*Proof.* For every  $\varepsilon$ , the deterministic mean in (2.2) is strictly increasing in  $s$ , so  $q^\varepsilon(\cdot, t)$  is nondecreasing. Proposition 5.1 gives uniform convergence near the target, while Theorem 4.1 makes the limiting curve continuous and strictly increasing. Standard monotone-quantile convergence then applies.  $\square$

This is the precise sense in which the finite deterministic system inherits the limiting knee: its endpoint probability curves and their quantile locations converge. No statement about first-passage times is needed.

## 6 Comparison with stochastic and Markov descriptions

The limit (3.2) is itself a parameter-dependent linear Markov diffusion. Consequently, the construction is not a no-go result against Markov models. A Markov model with the same  $s$ -dependent drift reproduces the limiting curve exactly.

What the deterministic model adds is a *source realization*: the Brownian forcing and its coefficient arise from a finite-dimensional invertible map plus an invariant ensemble. This distinction is useful:

- a fixed generator that contains no protocol coordinate cannot describe variation with that coordinate;
- a parameter-dependent Markov family can describe the variation directly;
- the source model explains one way such a family can emerge after hidden coordinates are discarded and scales are separated.

The parameter dependence is not created from nothing by projection. It is already present in the deterministic drift  $\alpha(s - s_0)$ . Projection and homogenization supply the effective diffusion law; the source protocol supplies the moving mean.

## 7 Relation to MTT and physical measurement

The model has the abstract pattern

$$\begin{array}{c} \text{invertible upper dynamics} \\ \downarrow \text{non-injective retained observable} \\ \text{effective stochastic law} \end{array}$$

That pattern is compatible with projection-first formulations such as MTT, but compatibility is not source selection. To promote this example into an MTT result, one would still need:

1. an explicit map from the selected MTT carrier and fixed-point dynamics to  $\mathbb{R} \times \mathbb{T}^2$ , or a replacement source system derived from that carrier;
2. a theorem identifying the physical retained observable with  $P(x, y) = x$ ;
3. geometric sources for  $\gamma, \alpha, s_0$  and the scale separation;
4. a physical record-forming interaction and its operational outcome algebra; and
5. a derivation of the relevant probabilities on that algebra.

None of those obligations is discharged by the present toy model. In particular:

- the invariant measure  $\mu$  is supplied as ensemble data;
- the binary threshold is chosen, not derived;
- the Gaussian sigmoid is not the Born rule for arbitrary quantum contexts;
- endpoint thresholding is not wave-function collapse; and
- no empirical knee location is predicted.

The result is nevertheless useful as a consistency example. It proves that finite-dimensional invertible source dynamics is not, by itself, incompatible with a diffusive projected limit or a sharply varying lower-level outcome frequency.

## 8 Status and conclusion

| Object                    | Status                                      |
|---------------------------|---------------------------------------------|
| Finite source map         | Explicit smooth diffeomorphism              |
| Projection                | Explicit and non-injective                  |
| Probability source        | Declared invariant ensemble $y_0 \sim \mu$  |
| Diffusion coefficient     | Exact, $\sigma^2 = 1/2$                     |
| Homogenization            | Imported rigorous theorem, specialized here |
| Threshold curve           | Exact fixed-time Gaussian law               |
| Finite-source convergence | Uniform on compact protocol intervals       |
| First-passage law         | Not claimed                                 |
| Quantum interpretation    | Born rule and collapse not derived          |
| MTT source selection      | Open                                        |

The corrected construction reaches a modest but exact conclusion. A smooth invertible finite-dimensional map, observed through a non-injective coordinate and equipped with an invariant hidden-state ensemble, can converge to an affine diffusion. Its fixed-time binary readout has an exact finite-width threshold curve, and the corresponding finite deterministic curves converge uniformly on compact protocol intervals.

The result should be used as an existence proof and a diagnostic model. It does not show that all abrupt phenomena are caused by projection, and it does not replace a physical source theorem. Its value is that every ingredient responsible for the lower stochastic law is now visible: invertible source dynamics, hidden-state ensemble, observational projection, scale separation, and an explicitly protocol-dependent drift.

## References

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